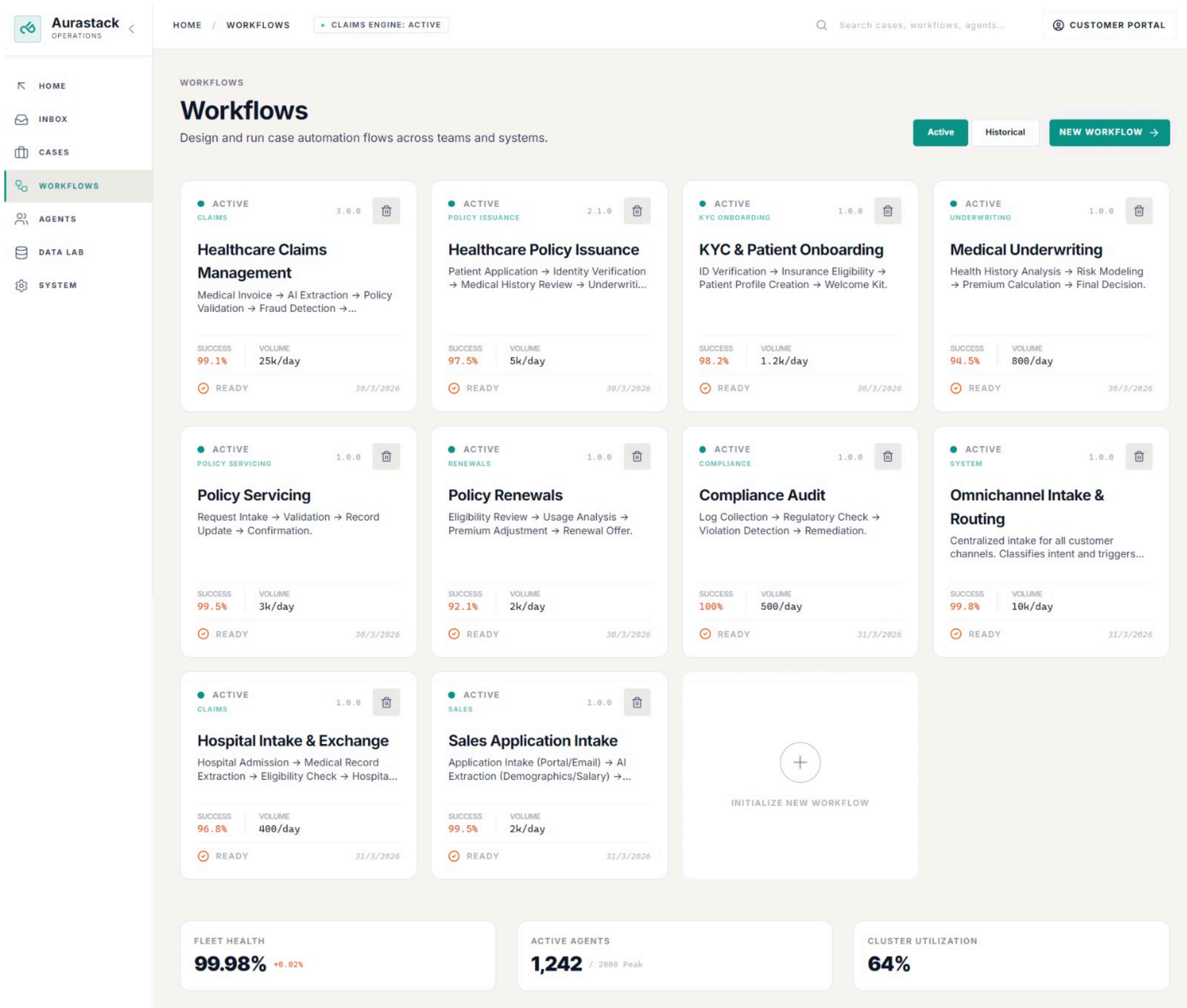
Workflows



Aurastack Workflows – Purpose-Driven System Description

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1. Healthcare Claims Management (claims_mgmt_v1)

Why This Exists

Claims are where insurance companies lose or make money. This workflow exists to **automate high-volume claims while protecting against fraud and leakage**.

Why It Is Important

Without this:

- Fraud slips through unnoticed
- Claims processing slows down
- Operational costs explode due to manual reviews

This workflow ensures **speed (STP) + control (fraud detection)** simultaneously.

Workflow Logic

This is our most complex "Straight-Through Processing" (STP) engine. It begins with FNOL (First Notice of Loss).

Core Components

- **Medical Data Extractor (doc_ext_01)**
  Extracts CPT codes with semantic understanding
- **Healthcare Fraud Sentinel (fraud_det_01)**
  Detects impossible or suspicious medical patterns

Decision Logic

Flags cases like repeated surgeries as "Impossible/Fraudulent"

Human-in-the-loop

If `fraud_score > 15`, case is paused and manually verified

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2. Healthcare Policy Issuance (policy_issuance_v1)

Why This Exists

Customer acquisition speed directly impacts revenue. This workflow exists to **compress underwriting + issuance into minutes instead of days**.

Why It Is Important

- Faster issuance = higher conversion rates
- Eliminates drop-offs during onboarding
- Reduces dependency on manual underwriting teams

Workflow Logic

Turns a lead into a policyholder in under 5 minutes

Core Components

- **Patient Identity Validator (kyc_val_01)**
- **Clinical Risk Modeler (risk_model)**

Decision Logic

If `risk_score < 30`, auto-generate and send policy

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3. KYC & Patient Onboarding (kyc_onboarding_v1)

Why This Exists

Every system needs a **clean, verified entry point for users**. This workflow ensures that **bad or duplicate identities don't enter the system**.

Why It Is Important

- Prevents fraud at the entry layer
- Ensures regulatory compliance
- Enables downstream workflows to trust the data

Workflow Logic

Lightweight onboarding for hospitals and minor products

Core Components

- API call to national insurance database

Output Actions

Creates profile + sends digital insurance card

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4. Medical Underwriting (medical_uw_v1)

Why This Exists

Not all customers are equal risk. This workflow exists to **price high-risk policies accurately without slowing down decision-making**.

Why It Is Important

- Prevents underpricing risky customers
- Protects long-term profitability
- Reduces dependency on expert underwriters

Workflow Logic

Used for high-value policies

Core Components

- **Medical Underwriter AI (med_analyst_01)**

Decision Logic

Highlights clinical risks

Premium Calculation

Applies loadings (e.g., smoker → +20%)

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5. Policy Servicing (policy_servicing_v1)

Why This Exists

After a policy is issued, customers continuously request changes. This workflow exists to **handle these safely without breaking system integrity**.

Why It Is Important

- Prevents invalid or unauthorized changes
- Maintains data consistency across systems
- Reduces customer support overhead

Workflow Logic

Handles post-sale lifecycle

Core Components

Validation rules (`policy_active`, `owner_verified`)

Execution

Writes updates back to core system

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6. Policy Renewals (policy_renewals_v1)

Why This Exists

Retention is more profitable than acquisition. This workflow exists to **ensure policies don't lapse and pricing stays aligned with risk**.

Why It Is Important

- Prevents revenue leakage from churn
- Adjusts pricing based on real usage
- Keeps portfolio profitable

Workflow Logic

Scheduled renewal system

Core Components

- **Renewal Risk Analyst (usage_analyst)**

Decision Logic

Suggests premium adjustment

Output

Automated renewal flow

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7. Compliance Audit (compliance_audit_v1)

Why This Exists

Insurance operates in a heavily regulated environment. This workflow exists to **continuously monitor and enforce compliance automatically**.

Why It Is Important

- Prevents legal penalties
- Ensures adherence to regulations (e.g., HIPAA)
- Builds trust with customers and regulators

Workflow Logic

Acts as internal monitoring system

Core Components

- Logs + compliance bot

Detection

Flags violations

Human-in-the-loop

Requires compliance officer approval

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8. Omnichannel Intake & Routing (omnichannel_routing_v1)

Why This Exists

All workflows need a **single intelligent entry point**. This workflow exists to **understand incoming requests and route them correctly**.

Why It Is Important

- Eliminates manual triaging
- Ensures faster response times
- Enables scalability across channels

Workflow Logic

Entry point of Insurance AI OS

Core Components

- **Aura Intake AI (router_agent)**

Decision Logic

Classifies intent and routes workflow

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9. Hospital Intake & Exchange (hospital_intake_v1)

Why This Exists

Incomplete data is one of the biggest blockers in healthcare workflows. This exists to **actively resolve missing information instead of stalling processes**.

Why It Is Important

- Prevents workflow deadlocks
- Reduces manual follow-ups
- Ensures continuity in processing

Workflow Logic

Handles missing admission data

Core Components

- **Hospital Exchange Coordinator (hospital_coord_01)**

Bidirectional Communication

Loops data back into workflow

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10. Sales Application Intake (sales_intake_v1)

Why This Exists

Sales pipelines need speed and instant feedback. This workflow exists to **convert applications into actionable outputs immediately**.

Why It Is Important

- Reduces lead drop-off
- Improves customer experience
- Connects sales and backend systems instantly

Workflow Logic

Lead-to-code engine

Core Components

- **Sales Intake Specialist (sales_intake_01)**

Integration

Calls external API

Output

Generates and delivers code instantly
